Air Displacement Plethysmography is an Accurate and Feasible Noninvasive Measure of Fat-Free Mass in Children With Intestinal Failure

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ABSTRACT

Background: The nutritional status of children with intestinal failure (IF) can be difficult to determine using body weight and currently available anthropometric techniques. Air displacement plethysmography (ADP) is a noninvasive measure of whole-body composition that measures body mass and volume, with a calculation of percent body fat (%BF) and fat-free mass (FFM) that may be useful during the provision of specialized nutrition.

Objectives: To evaluate the validity and feasibility of measuring body composition in children with IF using ADP compared with deuterium dilution (DD), as well as secondarily with other measures of body composition, namely bioelectrical impedance analysis (BIA), dual-energy X-ray absorptiometry (DXA), and four-site skinfold anthropometry.

Methods: We conducted a prospective cohort study of 18 children recruited through the Center for Advanced Intestinal Rehabilitation at Boston Children's Hospital. Patients 2–17 years of age with IF dependent on parenteral nutrition (PN) for more than 90 days were included. Spearman rank correlation and Bland-Altman limits of agreement (LOA) analysis were used to compare ADP to 4 alternative measures of body composition.

Results: Eighteen children with IF, median age 7.1 [interquartile range (IQR) 5.4–9.3] years, 9 female (50%), and median residual bowel length 31 (IQR 22–85) cm were enrolled. Median PN energy intake was 46 (IQR 39–49) kcal/kg/day. Incomplete bladder emptying lead to invalid measures of DD in 4 subjects. Spearman correlation coefficients for %BF were low to moderate between ADP and DD ($r = 0.29$), DXA ($r = 0.62$), BIA ($r = 0.50$), and skinfold ($r = 0.40$). Correlations for FFM were high between ADP and these other measures (range 0.95–0.98). Comparing ADP with DD and skinfold measures, Bland-Altman analysis showed small mean bias (−1.9 and +1.5 kg) and acceptable 95% LOA ranges (10.7 and 22.9 kg), respectively, with larger bias (−10.7 and −7.7 kg) and LOA ranges (38.7 and 45.2 kg) compared to DXA and BIA. %BF by ADP and skinfold thickness were moderately correlated ($r = 0.43$) with low bias (−0.2%) but very wide LOA (25.7%).

Conclusions: Body composition via ADP is feasible and valid in children with IF as a measure of FFM but appears less suitable for the measurement of %BF. The technique holds promise as a noninvasive measure of body composition to assess the efficacy of nutritional, medical, and surgical interventions.

Key Words: air displacement plethysmography, bioelectric impedance analysis, body composition, deuterium dilution, dual X-ray absorptiometry, four-site skinfold anthropometry, intestinal failure, parenteral nutrition, short bowel syndrome

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Children with intestinal failure (IF) often require parenteral nutrition (PN) as they attain enteral tolerance and autonomy (1,2), and the decision to wean or advance PN is often made on the basis of serial body weight and height changes. Weight gain, however, can be inaccurate in this population due to changes in hydration, abdominal distension (due to organomegaly or intestinal dilatation), or fluid shifts, making changes in muscle mass difficult to detect. Accurate and valid measures of body composition (fat and fat-free mass, or FFM) are more desirable metrics to follow.

Currently available body composition techniques include bedside measures of skinfolds, as well as bioelectrical impedance analysis (BIA), dual-energy X-ray absorptiometry (DXA), and the generally accepted gold standard of deuterium dilution (DD). We had previously evaluated BIA compared with these other techniques and found that BIA compared favorably with DD with

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What Is Known

- Body weight is a commonly used but sometimes problematic metric to evaluate the efficacy of nutrition support of children with intestinal failure (IF).
- Body composition is a physiologically more relevant metric, and new technologies have been developed to measure fat and fat-free mass (FFM) in children.

What Is New

- We compared 4 techniques of body composition in 18 children with IF dependent on parenteral nutrition.
- Air-displacement plethysmography performed well in measuring FFM but less well in measuring percent body fat, compared with the gold standard deuterium dilution and other techniques.

respect to measurement of lean body mass (LBM) but had wide limits of agreement (LOA) with respect to percent body fat (%BF) (3). When applying to pediatric IF, air displacement plethysmography (ADP) is a new technique that allows body density (and therefore body composition) to be estimated by determination of body volume (volume displaced of air, akin to underwater weighing) and body weight using age and sex-specific equations. ADP is quick and involves no exposure to radiation. The technique has been validated in animal models (4) and applied to healthy children (5–8) but has had limited application to children with chronic illnesses including IF (9,10). We therefore examined the feasibility and validity of body composition measurements in children with IF using ADP, and compared these results with those obtained from DD, BIA, DXA, and bedside anthropometric methods (sum of four-site skinfolds).

METHODS

We performed a cross-sectional study in children with IF followed at the Center for Advanced Intestinal Rehabilitation at Boston Children's Hospital (BCH). The study was registered as NCT02561819. IF was defined as congenital or acquired gastrointestinal disease with a history of at least 90 days of PN dependence. The study was conducted from February 2016 to March 2017.

Excluded were children younger than 2 years or older than 17 years, had a cardiac pacing device or other electrical device that might interfere with BIA or DXA, ongoing fluid imbalance, clinically evident shifts in fluid compartments (eg, edema or ascites) or required ongoing fluid resuscitation (defined as daily fluid intake more than 150% of maintenance or fluid boluses more than 20 mL/kg/day), pregnancy or currently taking rectal, enteral, or parenteral corticosteroids (with the exception of topical creams or ointments or inhaled steroids for asthma). Due to the requirement for lying still during this test and our program's clinical protocol, DXA was performed in children aged 5 years or older. Subjects underwent study procedures in the Experimental Therapeutics Unit (ETU), with the support of ETU staff, and in the Gastroenterology Procedure Unit at BCH.

For ADP, the BOD POD air displacement plethysmograph device (COSMED, Inc., Concord, CA) was used. It consists of a single structure with a seat that separates the front (test) and rear (reference) chambers. The door includes a transparent window that the patient can see out of (11). The patients were fasted (including water) and rested for 2 hours prior based on general guidelines from the manufacturer's instructions, and PN discontinued for at least 1 hour prior to the test too allow for the disconnection of the central venous catheter from tubing and intravenous bags of fluid or nutrition while in the "pod." If a gastrostomy tube was present, it was closed. The patients wore a form-fitting bathing suit and swim cap to avoid the effects of clothing and hair. The body weight of the patient was measured on an integrated electronic scale and total body volume was measured within the ADP test chamber by applying gas laws that relate pressure changes to volumes of air in the enclosed chamber. Thoracic gas volume was measured according to specific guidelines when able. The entire test time is about 5 minutes and the patient is inside the device for a total of 2 minutes, which includes 2 measurements which take approximately 50 seconds each. The 2 body volume measurements that were closest in agreement were used by the system software to calculate the average body volume. Body density was then computed from body mass and body volume and converted to percentage and absolute FM and FFM using sex-specific equations by Fomon et al (12) and others.

We performed DD as previously described (3). A sterile, pyrogen-free dose of 0.15 g/kg of deuterium-enriched water (^2H_2O) was administered via a feeding tube or orally (based on patient feeding status). For patients receiving tube feeds or an oral diet, enteral or oral nutrition was stopped 2 hours preceding isotope administration. Any female subject ≥12 years or of childbearing potential had baseline urine sample tested for pregnancy with a urinary HCG test. If pregnancy was detected, the subject was excluded from participating in the study and returned to clinic for confirmatory pregnancy testing and counseling. Urine samples were obtained at baseline and then at 4 hours ± 30 minutes after the administration of deuterium. Urine specimens were centrifuged and stored immediately in a −80°C freezer until the time of analysis. Isotope enrichment was obtained by gas isotope ratio mass spectrometry using validated protocols in a commercial lab (Metabolic Solutions, Inc., Nashua, NH). FFM was calculated from total body water (TBW) using a hydration factor of 0.73. Fat mass (FM) was defined as total body weight − FFM. %BF was calculated as FM/body weight × 100.

BIA measurements were obtained with subjects in the supine position, using a multifrequency impedance device (Bodystat Quadscan 4000, Bodystat, Inc., Tampa, FL). Current-injector electrodes were placed just below the phalangeal-metacarpal joint in the middle of the dorsal side of the right hand and below the metatarsal arch on the superior side of the right foot. Detector electrodes were placed on the posterior side of the right wrist, midline to the pisiform bone of the medial (fifth phalangeal) side with the wrist semi flexed. Impedance was measured with a multi-frequency bioelectrical impedance analyzer using 5, 50, 100, and 200 kHz at oscillating current. An undisclosed proprietary equation developed by the manufacturer calculated TBW using the impedance at 5 and 200 kHz, body weight, height, age, and gender (information provided by manufacturer). Estimates of TBW from BIA were converted to kg by using a conversion factor equivalent to the density of water at 36°C (0.9937 g/cm^3). Hence, TBW was a calculated variable from measured impedance values. LBM, FM, and %BF values are calculated using TBW and body weight measurements.

Anthropometric measurements, including skinfolds, weight, and height, and 24-hour oral, enteral, and parenteral dietary recall were recorded using standard devices and methods by 2 trained nutrition research staff dedicated to the study. An electronic digital scale (ScaleTronics, White Plains, NY) accurate to 0.1 kg was used to measure body weight (kg). Standing height (cm) was measured by a Harpenden stadiometer (Holtain Ltd., Crymych, UK) to the nearest 0.1 cm. Weight for age z score, weight for height z score, and height for age z scores were calculated using the CDC standards. Triceps, biceps, iliac, and subscapular skinfold thickness were measured to the nearest 0.2 mm using Lange skin calipers. Mid-arm

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circumference (MUAC) was measured to the nearest 0.1 cm using a flexible non-stretchable plastic tape. Mid-arm muscle area (MAMA) was calculated according to the equation. MAMA = (MUAC − 3.1416 × triceps skinfold)^2/ 4×3.1416. FM (kg) and %BF were calculated from published equations using 4 skinfold thickness measurements (13). For skinfold measurements, FFM was derived from published calculations as the difference between measured body weight and measured FM. All anthropometric measurements were performed in triplicate by 1 of 2 independent observers, and the average value of these 3 measurements were recorded.

A convenience sample of subjects with IF who had undergone DXA measurements for bone density assessment were also added for comparative purposes. DXA measurements were obtained in the anterior posterior supine position using a Hologic Discovery A (Hologic, Inc., Marlborough, MA) fan beam scanner generating X-rays at 2 energy levels (100 and 70 kV). The device uses the differential attenuation of the X-ray beam at these 2 energies to calculate the bone mineral content and soft tissue composition in the scanned region. A whole-body scan followed by a hip/spine scan was performed including measurements of bone density and body composition from the head to distal feet in the supine position. The scan included bone mass and body composition from the head to distal feet while in the supine position. Data were expressed as grams of fat (FM), grams of lean tissue mass (LBM), and %BF. For DXA, FM and FFM were both analyzed and measured directly.

The BCH institutional review board approved the study and subjects' parents or guardians gave written informed consent.

Statistical Analysis

The primary a priori comparison was body composition via ADP and DD; secondary outcomes were comparisons between ADP and other measures of body composition. Categorical data were tabulated using frequency and percentage, and continuous data described using median and interquartile range (IQR). Spearman rank correlation coefficients were calculated for the 2 primary study outcomes (%BF and FFM), comparing estimates from ADP with the 4 other body composition methods to assess consistency. Bland-Altman LOA analysis was also performed to examine the agreement between ADP and the 4 other methods for %BF and FFM. All analyses were conducted in SAS Version 9.4 (SAS Institute Inc., Cary, NC).

Although our goal enrollment was 25 subjects, the achieved sample size of 18 subjects was driven by the local availability of suitable participants, budgetary constraints, and was meant to approximate our previous study of 15 subjects on this topic (3).

RESULTS

Eighteen subjects with IF were enrolled in the study (Table 1). The median age of the cohort was 7.1 (IQR 5.4–9.3) years and nearly all subjects (17) were prepubescent; 9 (50%) were female. Midgut volvulus and gastroschisis accounted for 72% of all IF diagnoses, and median residual bowel length was 31 cm (IQR 22–85). All subjects were receiving at least some form of PN (Table 1, Supplemental Digital Content 1, *http://links.lww.com/MPG/D243*), with median PN energy intake 46 kcal/kg/day (IQR 39–49) with 47% (IQR 34–57) of total calories provided by enteral nutrition. Standard anthropometric assessment confirmed that the subjects were somewhat stunted but of normal body mass index (BMI). Median (IQR) weight for age, height for age, and BMI for age *z* scores were −0.29 (−1.28 to −0.06), −1.34 (−2.05 to −0.15), and 0.03 (−0.39 to 0.48, respectively.

All 18 subjects underwent successful ADP, BIA, and skinfold measurements (Table 1). Due to presumed incomplete bladder emptying, data from 4 subjects undergoing DD were judged to be invalid, so 14 subjects had isotope data available. Eight of our

TABLE 1. Baseline characteristics (n = 18)

	Statistic*
Subject characteristics	
Age at consent, y, range 3.2–15.8	7.1 (5.4–9.3)
Pre-pubescent	17 (94%)
Female sex	9 (50%)
Primary diagnosis (subjects may have >1)	
Midgut volvulus	9 (50%)
Gastroschisis	8 (44%)
Intestinal atresia (n = 1 ileal, n = 4 jejunal)	5 (28%)
NEC	3 (17%)
Malrotation	2 (11%)
Pseudo-obstruction	1 (6%)
Other†	5 (28%)
Residual small bowel length, cm (n = 17), range 0–464	31 (22–85)
Current stoma	1 (6%)
Ever bowel lengthening enteroplasty	5 (28%)
Ever tapering enteroplasty	5 (28%)
Ever Omegaven	6 (43%)
Serum citrulline (μmol/L), range 5–52	17.5 (13.0–26.0)
Months prior to consent that citrulline was obtained, range 0–18	10.8 (4.4–14.3)
Ileocecal valve present (among n = 15 known)	2 (13%)
Bowel in continuity	17 (94%)
Intact colon	18 (100%)
Colonic resection	14 (78%)
Body composition measures	
ADP data available	18 (100%)
Percent body fat, range 4.9–27.3	18.8 (12.6–21.4)
Fat-free mass, kg, range 10.56–38.27	16.92 (13.74–22.29)
Deuterium data available‡	14 (100%)
Percent body fat, range 3.1–26.6	13.2 (5.3–16.8)
Fat-free mass, kg, range 12.6–47.1	19.0 (14.8–26.8)
DXA data available§	8 (44%)
Percent body fat, range 25.5–36.3	32.6 (30.6–35.6)
Fat mass, kg, range 5.7–14.9	8.2 (6.8–10.1)
Fat-free mass, kg, range 12.0–33.3	11.1 (13.1–19.2)
BIA data available	18 (100%)
Percent body fat, range 11.2–41.2	26.1 (23.0–31.9)
Fat mass, kg, range 2.9–11.5	5.5 (4.3–6.9)
Fat-free mass, kg, range 7.7–38.4	14.9 (12.8–20.9)
Four-site skinfold anthropometry available	18 (100%)
Mean of 3 weights, range 13.1–48.6	19.5 (17.4–26.7)
Mean of 3 heights, range 95.3–158.1	112.0 (101.6–128.9)
BMI (kg/m^2), range 13.9–20.9	15.9 (15.2–17.3)
Percent body fat from 4-site skinfold measurement, range 8.0–27.3	15.2 (13.8–22.0)
Fat-free mass (kg) from 4-site skinfold measurement, range 12.1–37.9	16.2 (14.9–21.4)

ADP = air displacement plethysmography; BIA = bioelectrical impedance analysis; BMI = body mass index; NEC = necrotizing enterocolitis. *Median (interquartile range) or number (percent). †Intrauterine volvulus (2), hypoganglionosis (1), vanishing gastroschisis (1), perforated viscus (1). ‡Four subjects with incomplete bladder emptying were omitted. §Eight subjects aged ≥5 years underwent dual-energy X-ray absorptiometry (DXA).

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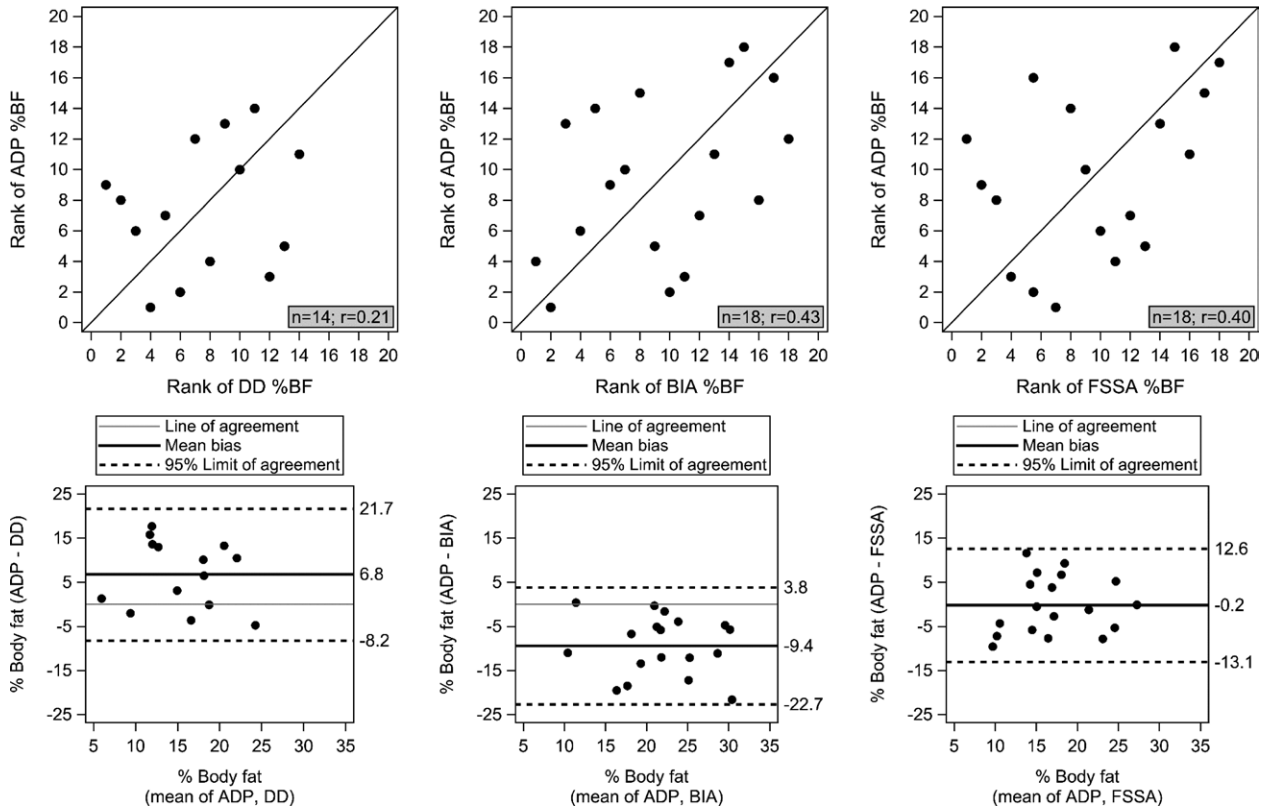

FIGURE 1. Spearman rank correlations (top row) and Bland-Altman (bottom row) limits of agreement plots for air displacement plethysmography (ADP) versus deuterium dilution (DD), bioelectric impedance analysis (BIA), and four-site skinfold anthropometry (FSSA) for percent body fat (%BF).

subjects had also undergone DXA measurements for bone density assessment so these data were also used for comparative purposes.

For the outcome %BF, Spearman rank correlation coefficients were low to moderate between ADP and DD (n = 14; $r = 0.21$), BIA (n = 18; $r = 0.43$), and skinfold thickness (n = 18; $r = 0.40$), reflecting only a modicum of consistency in the ranking of measurements compared to ADP (Fig. 1). Bland-Altman LOA analysis for ADP compared to these 3 comparator methods shows mean biases ranging from −9.4% to +6.8%, with very wide LOA. Spearman rank correlation coefficients (n = 8; $r = 0.54$) and LOA between %BF with ADP and DXA are shown in Figure 1, Supplemental Digital Content 2, *http://links.lww.com/MPG/D244*.

For FM, Spearman rank correlation coefficients (Fig. 2) were high between ADP and DD (n = 14; $r = 0.95$), and BIA (n = 18; $r = 0.97$) and skinfold thickness (n = 18; $r = 0.98$), reflecting high consistency in the ranking of these measurements compared to ADP. Bland-Altman LOA analysis for ADP compared to these 3 comparator methods showed mean biases ranging from −7.7 to +1.5 kg, with narrow LOA. ADP tended to underestimate FFM by an average of 1.9 kg compared with DD and overestimate FFM by an average of 1.5 kg compared with skinfold measures. Spearman rank correlation coefficients (n = 8; $r = 0.98$) and LOA between FFM with ADP and DXA are shown in Figure 1, Supplemental Digital Content 2, *http://links.lww.com/MPG/D244*.

DISCUSSION

We performed multiple measures of body composition in a cohort of children with PN-dependent IF. While our primary aim was to validate ADP against the reference standard DD technique, the fact that 4 subjects had incomplete bladder emptying and invalid deuterium outcomes called into question our use of deuterium in this population. We found that although there was low to moderate correlation and agreement between ADP measures of %BF with these comparative measures, the ADP measurement of FFM was well correlated with these comparators and had reasonable LOA.

Studies reporting comparative data using different body composition methodologies in IF have been limited. Jones et al (10) studied 9 adults with IF who underwent ADP testing as well as comparative measures with BIA and ultrasound. These authors expressed the relationship between methods via the intraclass correlation coefficient (ICC) (we used the Spearman rank correlation coefficient) and did not perform Bland-Altman analysis, which make direct comparisons of the 2 studies difficult. However, similar to our findings, the ICC they determined for FFM was higher for ADP versus BIA or ultrasound (correlations were 0.79, 0.66, respectively) than for FM (where correlations were 0.66, −0.005, respectively). An explanation for diversion between ADP and other modalities of body fat analysis may be the presence of increased intraluminal gas in patients with short bowel syndrome.

In a cohort study of 20 children with IF and PN-dependency, Neelis et al (9) described body composition using ADP. Their cohort had similar anthropometric characteristics to our subjects, including median weight for age and height for age z scores of −0.71 and −1.55, respectively, but more of their participants (57%) had functional IF such as pseudo-obstruction versus anatomic etiologies. These authors expressed their body composition data not in absolute kilograms or percentage but in fat-mass and FFM z scores, as calculated using published UK references. They reported median fat-mass and fat-free z scores of 0.80 and −1.53, respectively, which is suggestive of maintained FM but lower than average FFM in this cohort. Because of their analytic methods, direct comparison with

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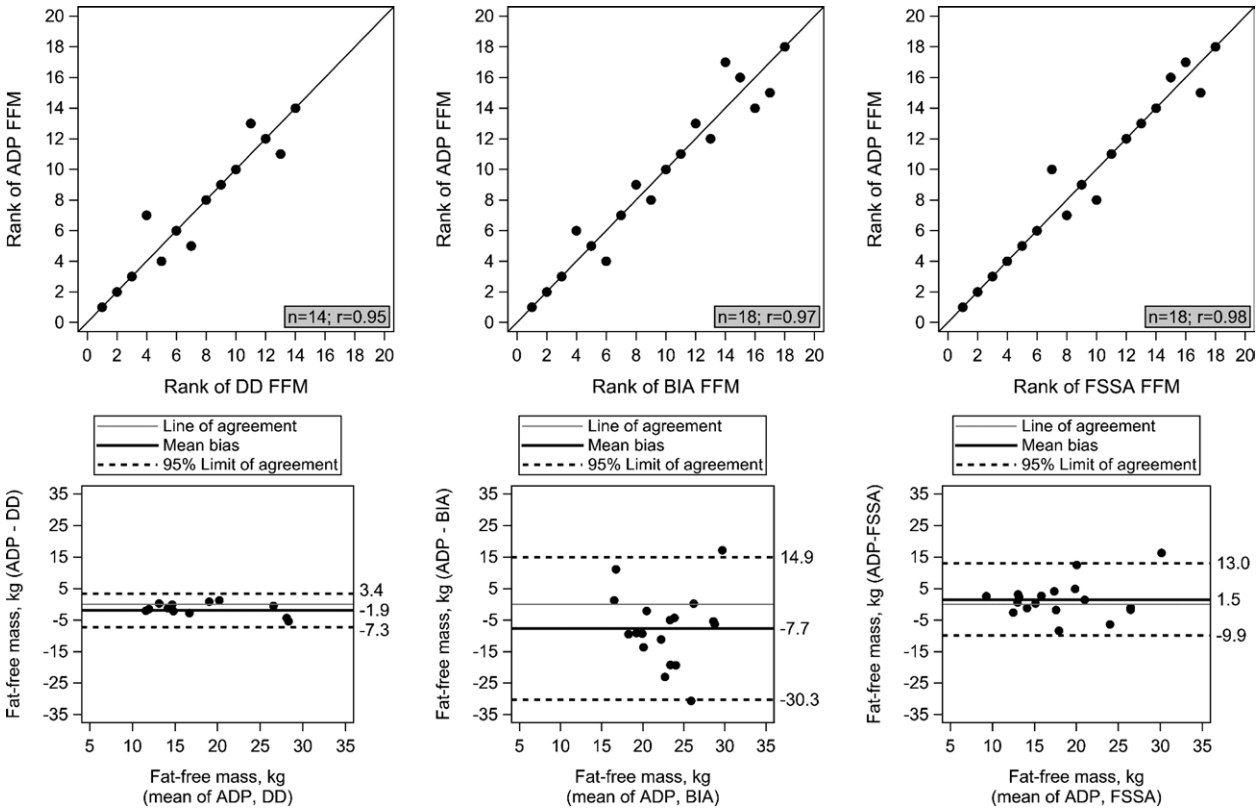

FIGURE 2. Spearman rank correlations (top row) and Bland-Altman (bottom row) limits of agreement plots for air displacement plethysmography (ADP) versus deuterium dilution (DD), bioelectric impedance analysis (BIA), and four-site skinfold anthropometry (FSSA) for fat-free mass (FFM).

our data is not possible, and comparison with other body composition methods was not performed.

Strengths of our study include the prospective nature of data collection, detailed descriptions of the nutritional status and intake of the participants, and the use of multiple comparator body composition methods (DD, DXA, BIA, and skinfold thickness) to ADP. Limitations include the clamping of gastrostomy tubes while in the "pod" which may have had an effect on measurements, the lack of valid deuterium measures in 4 subjects, limited sample size, and the cross-sectional nature of the study. Cost and access to ADP outside of academic settings are also limiting factors, both of which may impact clinical utility.

CONCLUSIONS

In summary, ADP is a feasible noninvasive and reasonably valid measure of FFM in young children with IF. Measures of FFM seem better assessed by the technique than %BF. As new nutritional, medical, and surgical interventions become available for the children with this condition, and as younger patients survive into adulthood, determinants of body composition, including longitudinal assessments, will be increasingly important outcomes for patients and clinical trials alike.

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